Alessandro Gaggioli 0000914713

FINAL REPORT

The system

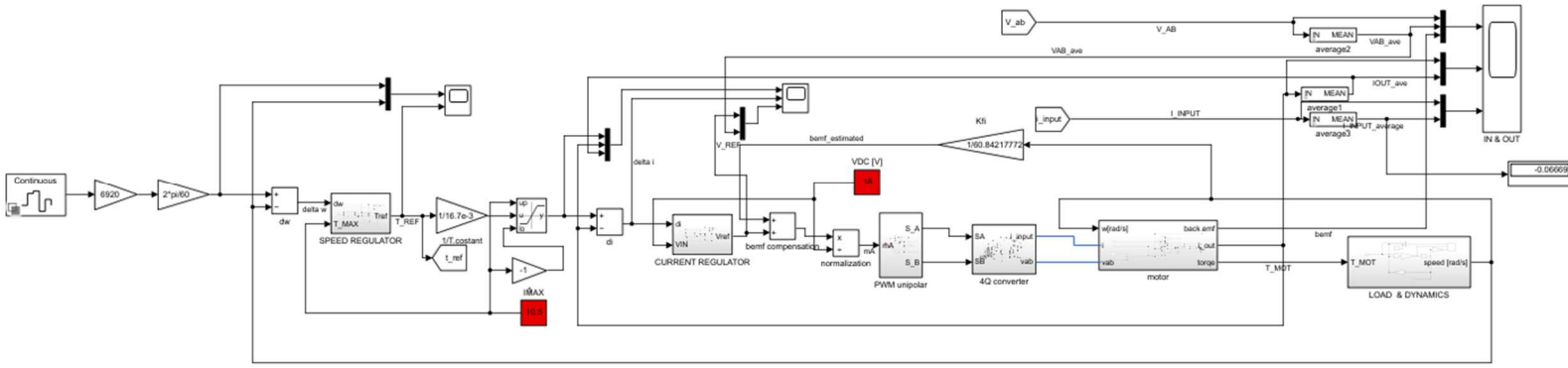
```
%Motor data
omega_nominal = 6370; %[rpm] nominal speed
Rw = 0.117; %[Ohm] winding resistance
Lw = 0.0245e-3; % [H] winding inductance
Kt = 16.7e-3; %[Nm/A] torque constant
Kspeed = 581; %[rpm/V] speed constant
J_m = 139e-7; %[kgcm^2] rotor inertia

%Load data
C_r_slope = 7.121e-5; %[Nm*s] load torque slope
J_r = 167e-7; %[kgcm^2] load inertia

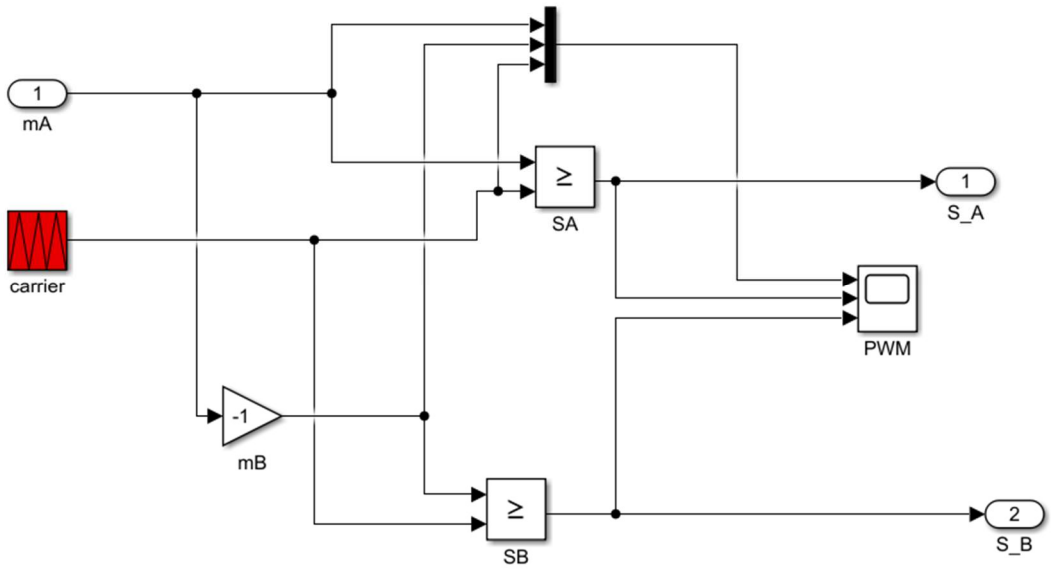
%Supply system
V_in = 18; %[V] supply voltage
R_s = 0.18; %[Ohm] source internal resistance
L_s = 0.18e-3; %[H] source internal inductance

%Converter
i_sat = 10.5; %[A] max output current
```

Simulink model

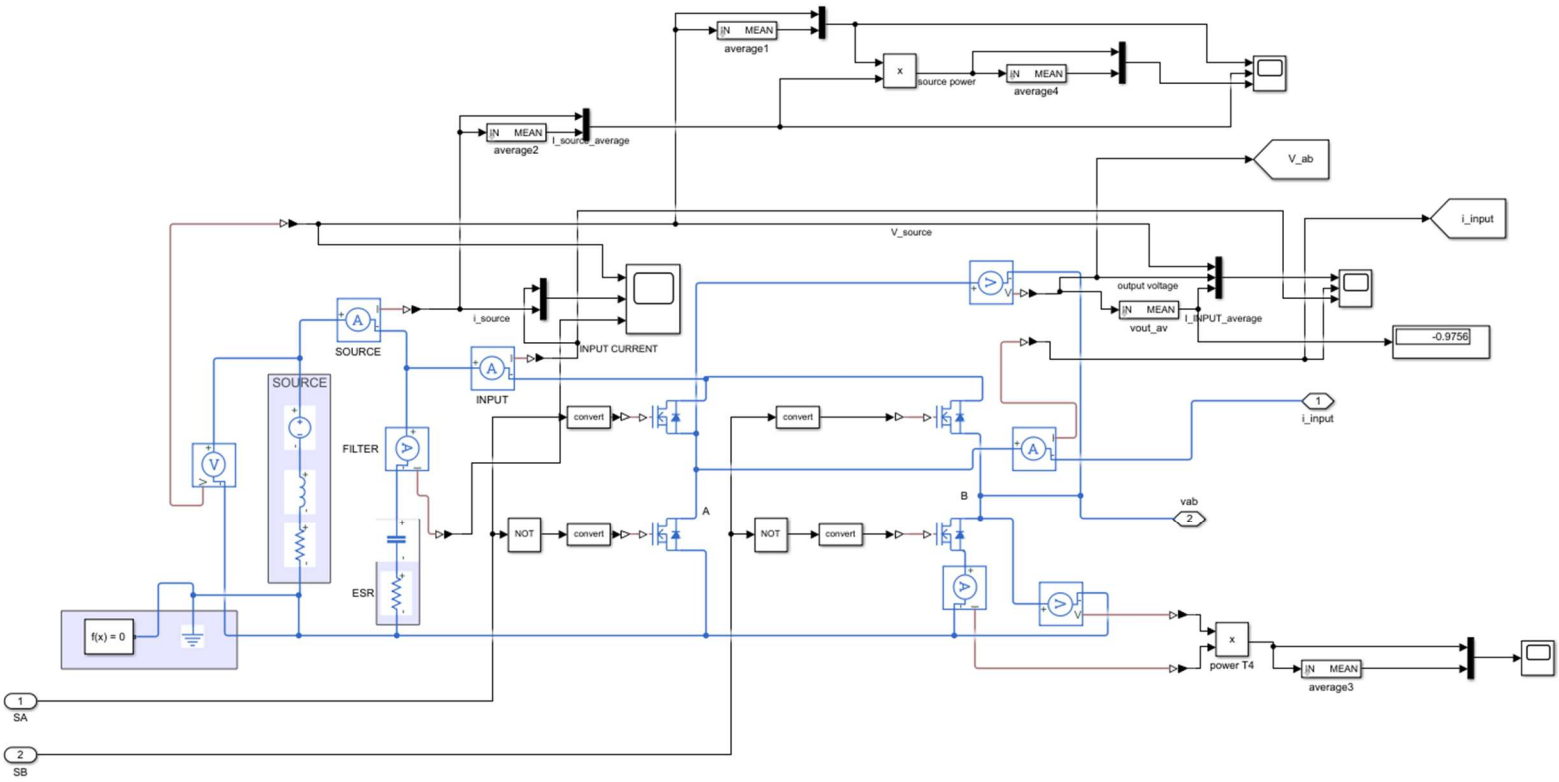


PWM model :

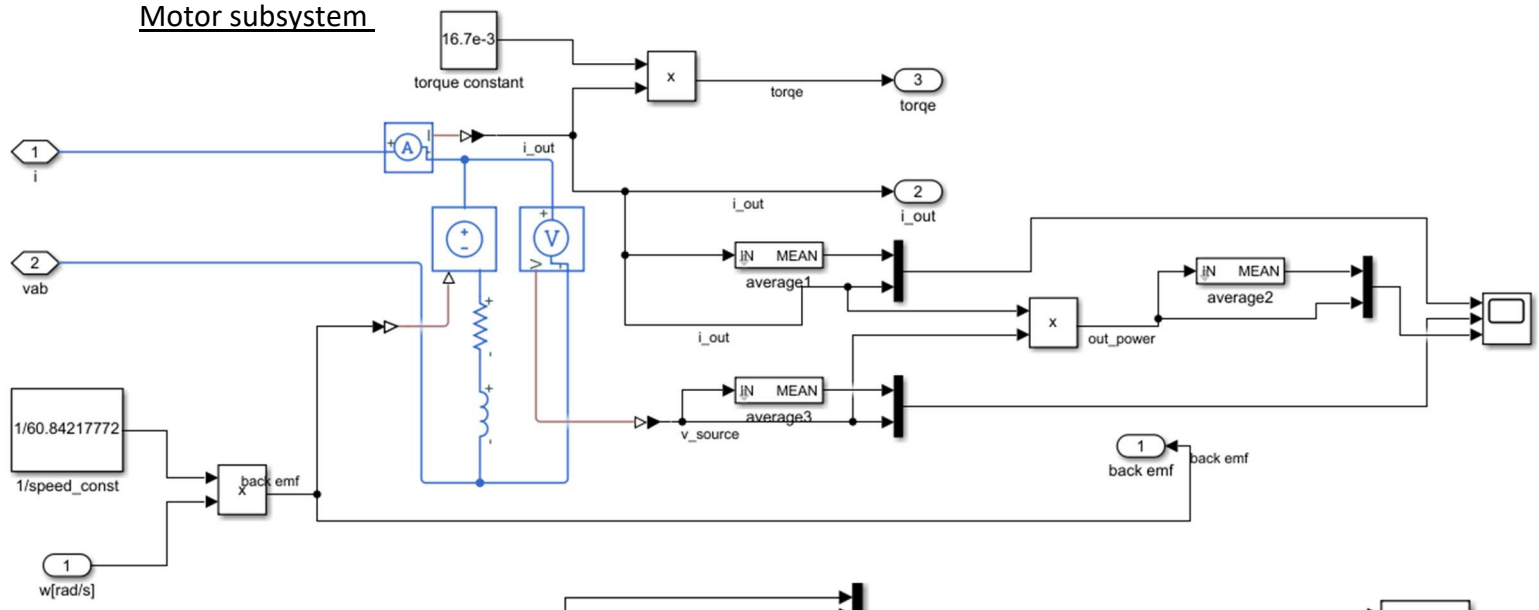


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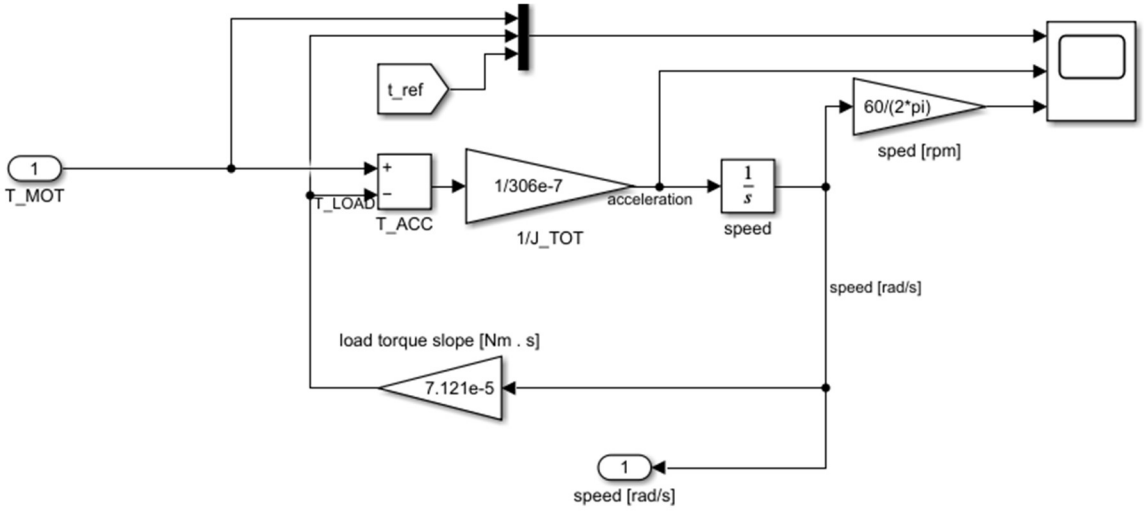
Real 4 quadrant DC/DC converter subsystem



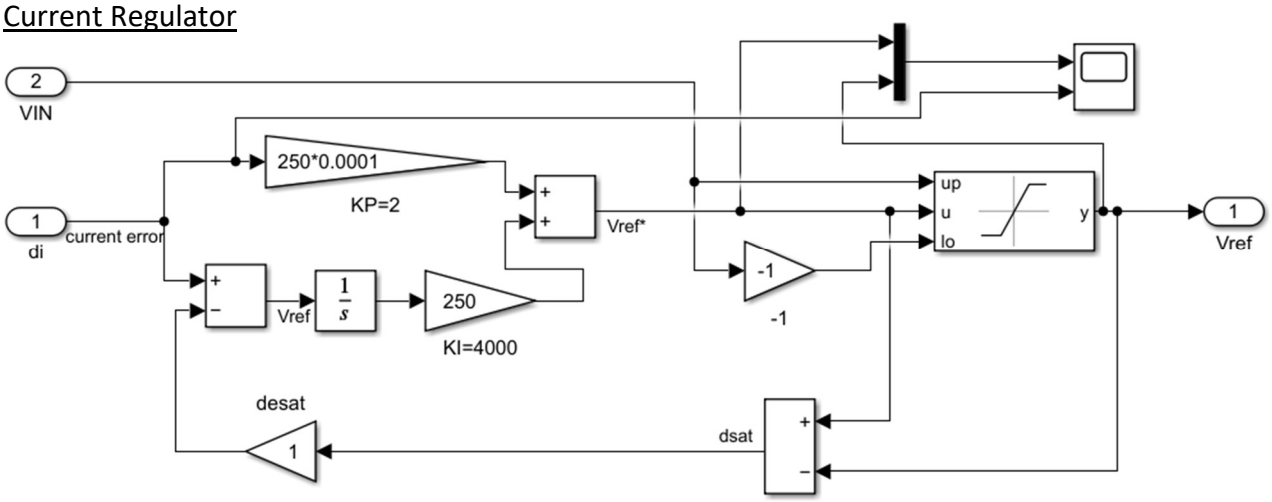
Motor subsystem



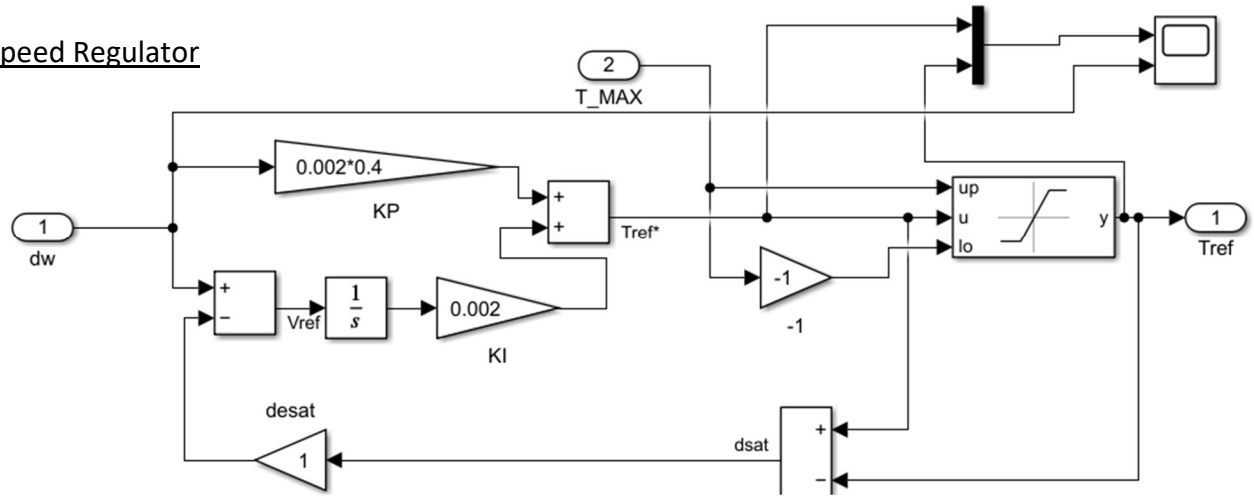
Load and Dynamics subsystem



Current Regulator



Speed Regulator



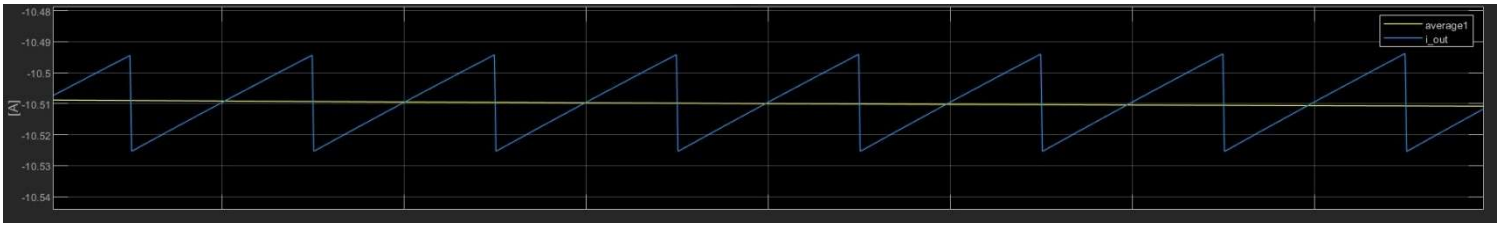
Parameters of the control scheme

Carrier frequency	100	kHz
Input filter	0.01	F
Max current saturation	10.5	A
KI current regulator	250	-
KP current regulator	250*0.0001	-
KI speed regulator	0.002	-
KP speed regulator	0.002*0.4	-

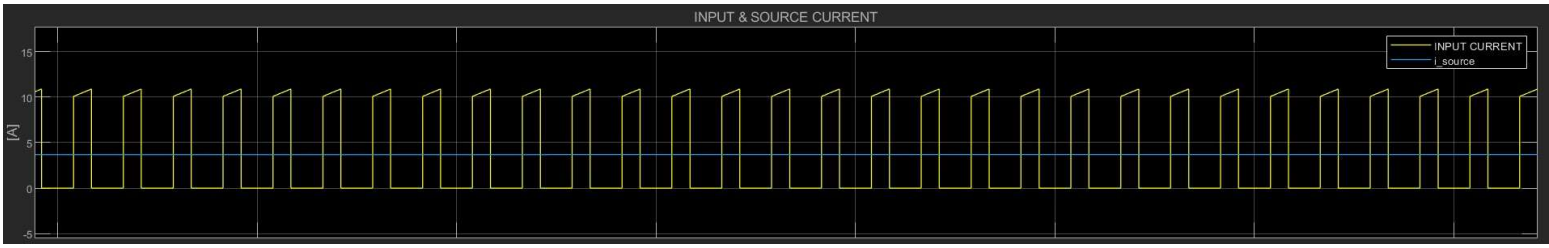
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Diagrams

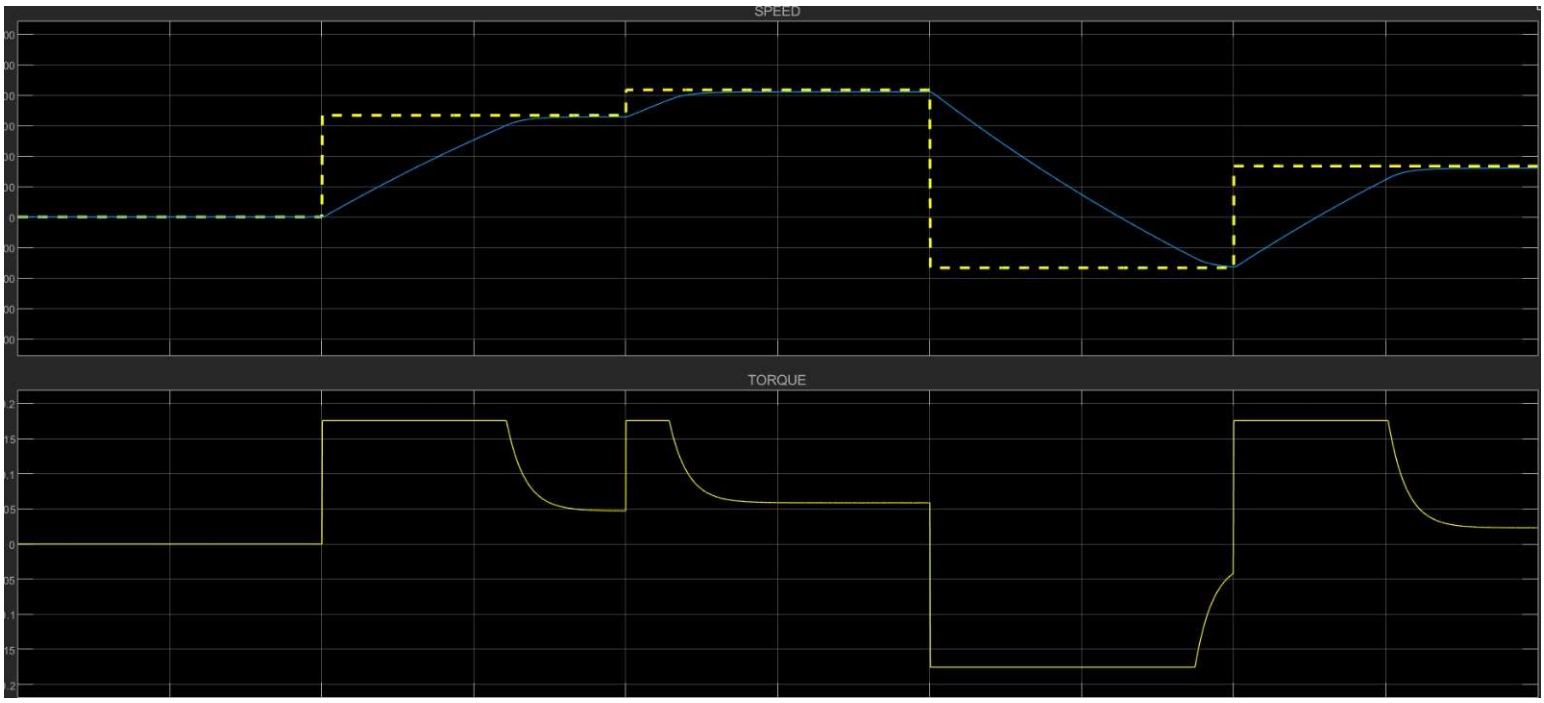
Output current ripple



Input current ripple



Speed and Torque



Final Results

Output current ripple	0.28%
Input current ripple	0.38%
Speed overshoot	0%
Rise time <100ms	not achieved

The rise time is an unachievable performance. Even if we try to turn up the integral gain, we only notice an overshoot of the nominal speed but the slope of the curve always remains the same .